

NIKITHA KUNAPAREDDY

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EDUCATION

B.Tech (Computer Science) , SRM AP University (2023 – Present)	CGPA: 9.13/10
AP EAMCET – State Rank: 616	
Class 12 , Narayana Junior College (2023)	96.3%
Class 10 , Narayana School (2021)	100%

WORK EXPERIENCE

CultureVO <i>Software Engineer Intern (Machine Learning & Generative AI)</i>	May 2025 – Aug 2025
- Engineered LLM-powered conversational AI, multimodal, and RAG systems serving 1000+ daily users, building applications such as Aura LLM Platform, Tiny Agents, ImageInsightLLM, and Symphony Summary Orchestra. - Developed scalable AI microservices using FastAPI, Redis, RabbitMQ, and Google Cloud Run, enabling sub-100ms retrieval and processing 10,000+ asynchronous tasks daily. - Implemented Prometheus-Grafana observability and CI/CD pipelines, achieving 99.9% uptime, reducing P95 latency by 30%, and improving production reliability.	

PROJECTS

MultiMind-RAG-Agent: AI-Driven Knowledge Orchestration Platform | *RAG, Vector Embeddings, LLM, Redis, RabbitMQ*

- Developed a real-time **knowledge orchestration platform** unifying dynamic document analysis for **10,000+ documents** using **Vector Embeddings**, conversational memory and live web research into an adaptive AI interface.
- Engineered **sub-50ms memory retrieval** using **Redis**, processing **5,000+ asynchronous tasks daily** with **RabbitMQ**, and personalized retrieval using custom **RFM scoring algorithms**.
- Integrated advanced **Retrieval-Augmented Generation (RAG)** pipelines, improving answer relevance by **30%** and delivering personalized, context-aware insights for research and decision-making at scale.

Adaptive-Photo-Restoration-Pipeline | *Python, OpenCV, NumPy, Scikit-Image, CNN*

- Developed an adaptive photo restoration pipeline for degraded historical images using **denoising**, **inpainting**, and **contrast enhancement**, achieving **20.15 dB PSNR** and **0.83 SSIM** without GPU acceleration.
- Designed a noise analysis framework using **2D FFT analysis** and a lightweight **3-layer CNN** to classify multiple noise types and improve **BRISQUE scores by 44–47%**.
- Implemented **Hough Transform**-based fold detection with **Telea Fast Marching inpainting** and optimized image enhancement parameters for improved restoration quality.

DealPulse AI: Smart Opportunity Stage Predictor | *Python, PyTorch, Transformers, FastAPI, MLflow, SHAP*

- Built an AI-powered **CRM opportunity intelligence system** to automatically predict sales deal stages from unstructured **CRM conversation notes**, reducing manual pipeline updates.
- Fine-tuned a **DistilBERT** model on **CRM notes data** using an end-to-end NLP pipeline with preprocessing, tokenization, label encoding, and model optimization.
- Deployed a **FastAPI** service with **MLflow** tracking and **SHAP explainability**, providing real-time predictions, confidence scores, and key **CRM insights**.

Familia: Cross-Cultural Human Connection Platform | *Next.js, FastAPI, Supabase, PostgreSQL, WebSockets*

- Developed a scalable **cross-cultural social networking platform** with AI-powered multilingual communication, voice/video-based verification, and real-time messaging for global user interaction.
- Engineered a secure full-stack architecture using **Next.js**, **FastAPI**, **Supabase**, and **PostgreSQL** with **WebSockets**, JWT authentication, and Row-Level Security (RLS) for low-latency communication.
- Built interactive engagement features including relationship progression levels, global family rooms, leaderboards, and social activities to enhance long-term community interaction.

TECHNICAL SKILLS

Languages & Core: C, C++, Java, Python, JavaScript, PHP, SQL, Data Structures & Algorithms

Frontend & Backend Architectures: HTML5, CSS, FastAPI, Flask, Streamlit, RESTful APIs

Databases & Cloud Systems: MySQL, PostgreSQL (Supabase), CI/CD Pipelines, Google Cloud Platform (Cloud Run)

AI & Machine Learning: Deep Learning, Generative AI, LLMs, RAG Pipelines, Conversational Agents, OpenCV, Transformers, Real-Time Inference

DevOps & Version Control: Git, GitHub

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Certified Java Programmer** (Oracle)
- Generative AI Certification** (LinkedIn Learning)